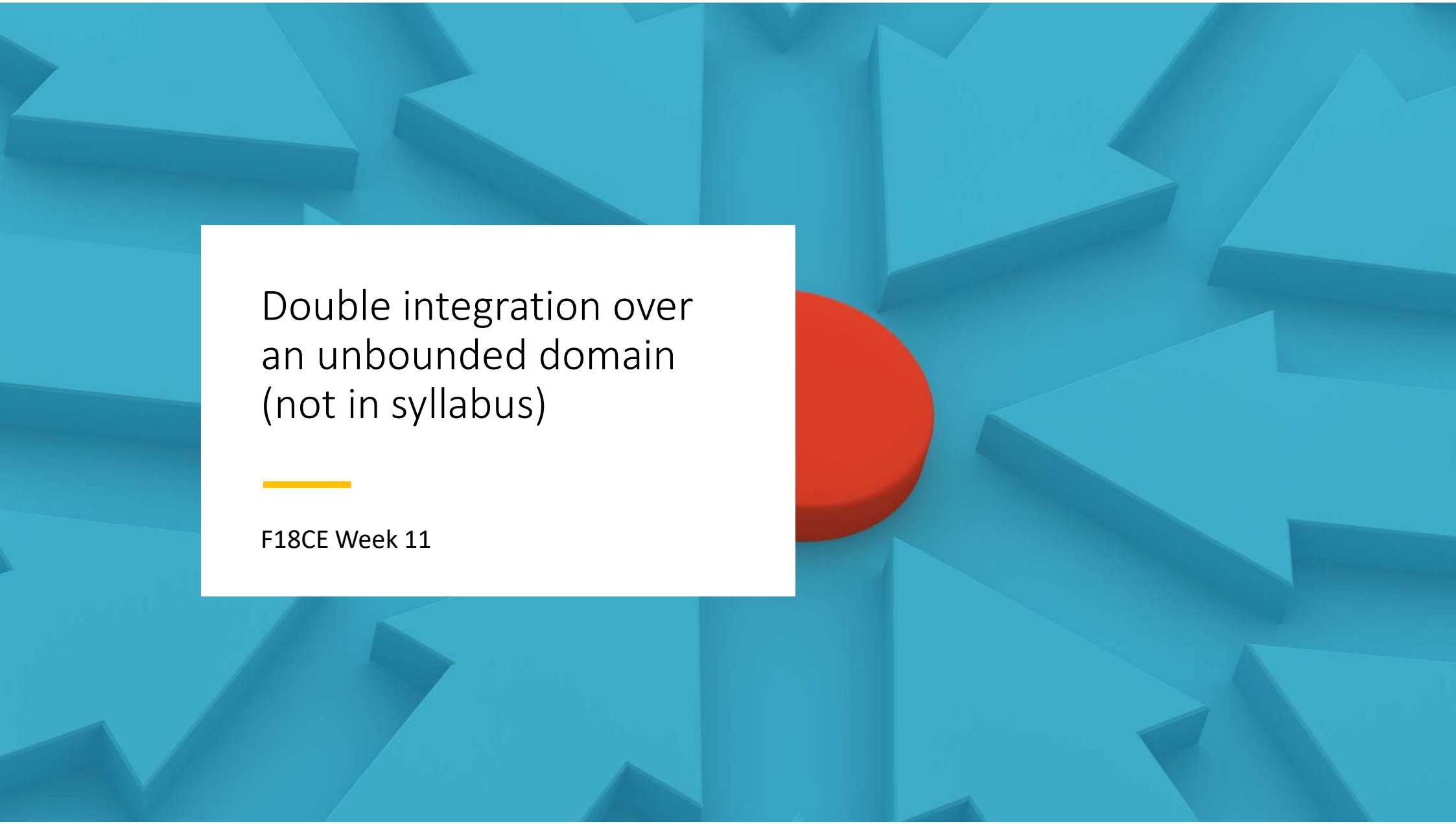


Double integration over an unbounded domain (not in syllabus)

F18CE Week 11

A Question

Suppose we try to find the area under the curve for function $y = x^{-2}$ from $x = -1$ to $x = 1$, then, if we integrate directly without thinking much, we will obtain

$$\int_{-1}^{1} \frac{1}{x^2}\, dx = \left[-\frac{1}{x} \right]_{-1}^{1} = 2\,.$$

However, this is incorrect! There is a discontinuity at point $x = 0$.

What should we do when encounter this scenario?

First, we should test whether $\int_{-1}^{1} \frac{1}{x^2}\,dx$ converges and if the answer is yes we can proceed to determine its value.

$$
\begin{aligned}
\int_{-1}^{1} \frac{1}{x^2}\,dx &= \lim_{a\to 0^-} \int_{-1}^{a} \frac{1}{x^2}\,dx + \lim_{b\to 0^+} \int_{b}^{1} \frac{1}{x^2}\,dx \\
&= \lim_{a\to 0^-} \left[-\frac{1}{x}\right]_{-1}^{a} + \lim_{b\to 0^+} \left[-\frac{1}{x}\right]_{b}^{1} \\
&= -\lim_{a\to 0^-} \left(\frac{1}{a} + 1\right) - \lim_{b\to 0^+} \left(1 - \frac{1}{b}\right) \\
&= -(-\infty + 1) - (1 - \infty) \\
&= \infty
\end{aligned}
$$

Here we see that $\int_{-1}^{1} \frac{1}{x^2}\,dx$ diverges.

Definition 10.1 Double Integral over unbounded domain

Consider an unbounded region $D \subset \mathbb{R}^2$ and a function $f(x,y) : \mathbb{R}^2 \to \mathbb{R}$ defined on D.

Let $I := \iint_D f(x,y)\,dxdy$ and we say that the double integral exists if

1. f is integrable on $D \cap D_R$ for any expanding sequence of bounded regions D_R,
2. $\lim_{R\to\infty} \iint_{D\cap D_R} f(x,y)\,dxdy$ exists, and
3. $I = \lim_{R\to\infty} \iint_{D\cap D_R} f(x,y)\,dxdy\,.$

Remark

1. f is continuous on D, then f is integrable on D.
2. The value of I is independent of the choice of D_R .

Theorem 10.2

Let D be an unbounded region on the plane and let $f(x,y) \geq 0$ be defined on D. Given a family of expanding bounded regions D_R, if

$$I = \lim_{R \to \infty} \iint_{D \cap D_R} f(x,y)\, dxdy$$

exists then $\iint_D f(x,y)\, dxdy$ exists and is equal to I.

Theorem 10.3

Given an unbounded region D and a function $f(x,y) : \mathbb{R}^2 \to \mathbb{R}$ defined on D then

$$\iint_D f(x,y)\,dxdy \text{ converges if and only if } \iint_D \left|f(x,y)\right| dxdy \text{ converges.}$$

Example 10.4

Let $D = \left\{(x, y) \in \mathbb{R}^2 : x, y \geq 1\right\}$ and calculate $\iint_D (xy)^{-2}\, dxdy$.

Solution. Let $D_R = \left\{(x,y) \in \mathbb{R}^2 : 1 \le x, y \le R\right\}$. Then $f(x,y) = (xy)^{-2} \ge 0$ (only need to consider one region) is continuous on $D \cap D_R$ $(= D_R)$, so f is integrable on $D \cap D_R$. Next, we determine whether $\lim\limits_{R\to\infty} \iint\limits_{D\cap D_R} f(x,y)\,dxdy$ exists or not.

$$\lim_{R\to\infty}\iint_{D\cap D_R} f(x,y)\,dxdy \;= \lim_{R\to\infty}\iint_{D_R}(xy)^{-2}\,dxdy = \lim_{R\to\infty}\left(\int_1^R x^{-2}dx\right)^2 = \lim_{R\to\infty}(R^{-1}-1)^2 = 1$$

We see that the limit exists. Furthermore, the limit is equal to 1. Therefore,

$$\iint_D (xy)^{-2}\,dxdy \;= 1.$$

Example 10.5

Let $D = \left\{(x, y) \in \mathbb{R}^2 : y \geq 0\right\}$. Calculate $\iint\limits_D e^{-(x^2+y^2)}\, dxdy$.

Solution. The occurrence of $x^2 + y^2$ prompted us to make substitution of $x = r\cos\theta$ and $y = r\sin\theta$. As such, we will consider a bounded area D_R which is a semicircle. i.e., $D_R = \left\{(x, y) \in \mathbb{R} \times \mathbb{R}^{\#} : x^2 + y^2 \leq R^2\right\}$ (where $\mathbb{R}^{\#}$ = set of non-negative real numbers). With the substitution of $x = r\cos\theta$ and $y = r\sin\theta$, and $0 \leq r \leq R$ and $0 \leq \theta \leq \pi$. We see that $f(x, y) = e^{-(x^2+y^2)} \geq 0$ and f is continuous on $D \cap D_R$ $(= D_R)$.

$$\begin{aligned}
\lim_{R\to\infty}\iint_{D\cap D_R} f(x,y)\,dxdy &= \lim_{R\to\infty}\iint_{D_R} e^{-x^2-y^2}\,dxdy \\
&= \lim_{R\to\infty}\int_0^{\pi}\int_0^{R} e^{-r^2}\,r\,drd\theta \\
&= \lim_{R\to\infty}\int_0^{\pi} d\theta\int_0^{R} re^{-r^2}\,dr \\
&= \tfrac{1}{2}\pi\lim_{R\to\infty}(1-e^{-R^2}) = \tfrac{1}{2}\pi
\end{aligned}$$

limit exists. Furthermore, the limit is equal to $\pi/2$. Therefore,

$$\iint_D e^{-(x^2+y^2)}\,dxdy = \pi/2.$$

Remark

The computation of $\iint_D f(x,y)\,dxdy$ when D is unbounded may not always be as easy as shown on two examples above. Some numerical methods can be explored to solve this problem to find an estimation of the answer but before we proceed, we should ask: is the double integral converges?

To answer this, we need to test the convergent of the integral. In particular, we can make use of the following theorem.

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Theorem 10.6 Comparison Test

Let D be an unbounded region on the plane and let $f(x,y), g(x,y) : \mathbb{R}^2 \to \mathbb{R}$ be two functions defined on D such that each function is integrable on any $D_R \cap D$. Suppose further that for some $c > 0$.

$$0 \leq \left|g(x,y)\right| \leq \left|f(x,y)\right| \; \forall (x,y) \in D \text{ and } \sqrt{x^2 + y^2} \geq c$$

then the following two implications hold

(a) f integrable on $D \Rightarrow$ g integrable on D

(b) g not integrable on $D \Rightarrow$ f not integrable on D

Example 10.7

$$\text{Let } D = \left\{(x,y) \in \mathbb{R}^2 : x, y \geq 1\right\}. \text{ Then } \iint_D \frac{\sin(x^2 y)}{x^2 y^2 + \cos^2(xy)}\, dxdy \text{ converges}$$

Proof. Note that $\left|\dfrac{\sin(x^2y)}{x^2y^2+\cos^2(xy)}\right| = \dfrac{\left|\sin(x^2y)\right|}{x^2y^2+\cos^2(xy)} \leq \dfrac{1}{x^2y^2+\cos^2(xy)} \leq \dfrac{1}{x^2y^2}$ since $0 \leq \cos^2(xy) \leq 1 \,\&\, x^2y^2 \geq 0 \Rightarrow x^2y^2+\cos^2(xy) \geq x^2y^2 \geq 0$. As shown in Example 10.4, $\iint\limits_D (xy)^{-2}\,dxdy = 1$, hence by comparison test we obtain the result.

Definition 10.8 Bounded function in $\mathbb{R}^2$

A function of two variables $f(x,y)$ is called bounded on a bounded region $D \subset \mathbb{R}^2$ if there exists a number K such that $\left|f(x,y)\right| \leq K$ for all $(x,y) \in D$. If such a number does not exist, then we say that f is unbounded on D.

Example 10.9

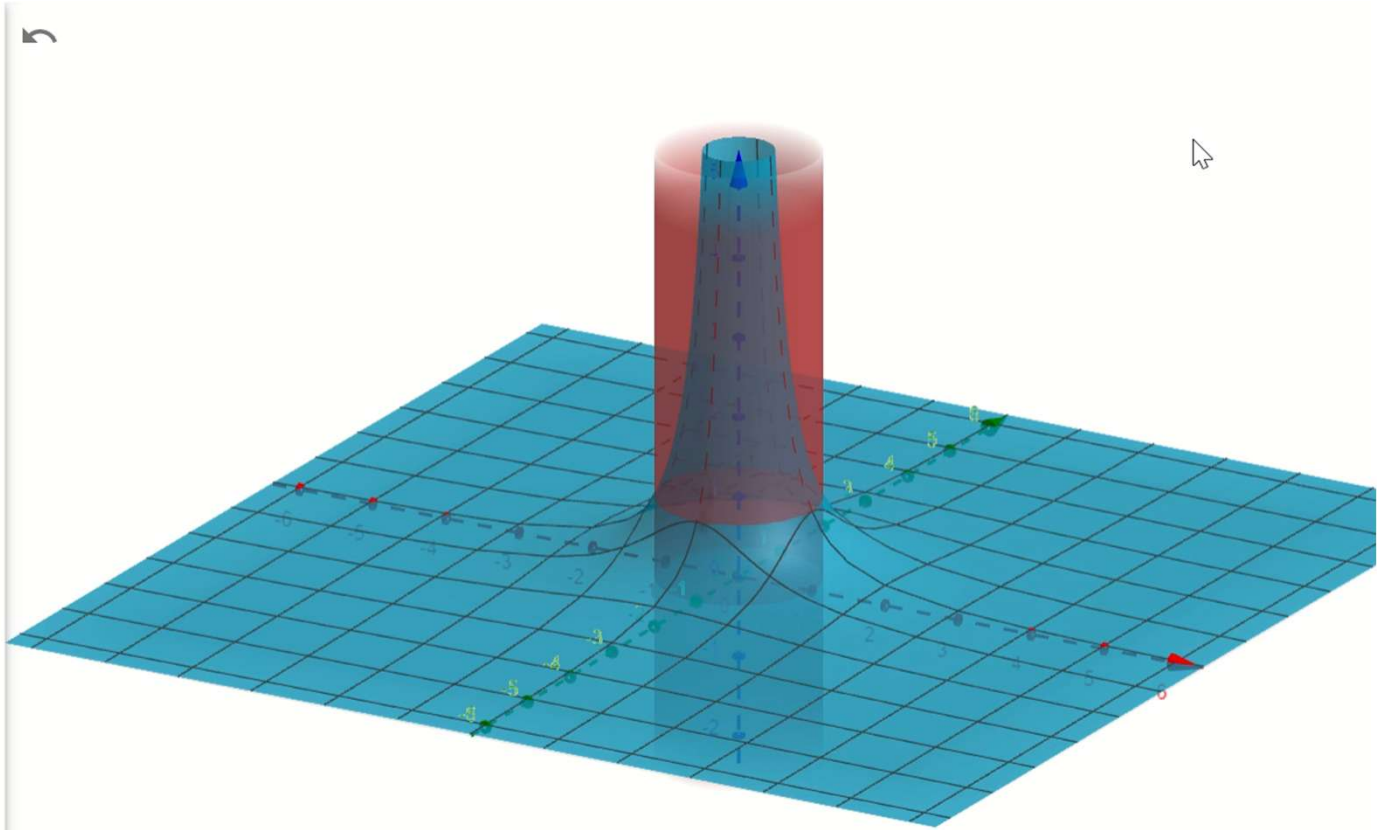
$$f(x,y) = \begin{cases} \frac{1}{x^2+y^2}, & (x,y) \neq 0 \\ 1, & (x,y) = 0 \end{cases}$$

considered on the unit disc B_1 is well defined but is unbounded. The origin is called a singular point of this function because f is unbounded on any neighbourhood (small disc) containing the origin and is bounded outside of such neighbourhood.

$a : z = \frac{1}{x^2 + y^2}$

eq1: $x^2 + y^2 = 1$

Input...



GeoGebra 3D Calculator

Remark

Let $f(x,y) \geq 0$ be a function defined on a region D. We can choose a region $D_R \subset D$ such that D_R does not consists of singular points in D and define

$$\iint\limits_{D} f(x,y)\,dA = \lim_{R\to 0} \iint\limits_{D_R} f(x,y)\,dA\,.$$

The convergence can be verified if the limit above exists for one particular choice of D_R.

Example 10.10

Let $f = (xy)^{-1}$ be defined on an open square $D = \{(x, y) : 0 < x, y < 1\}$.

Determine the convergence of $\iint_D (xy)^{-1}\, dA$.

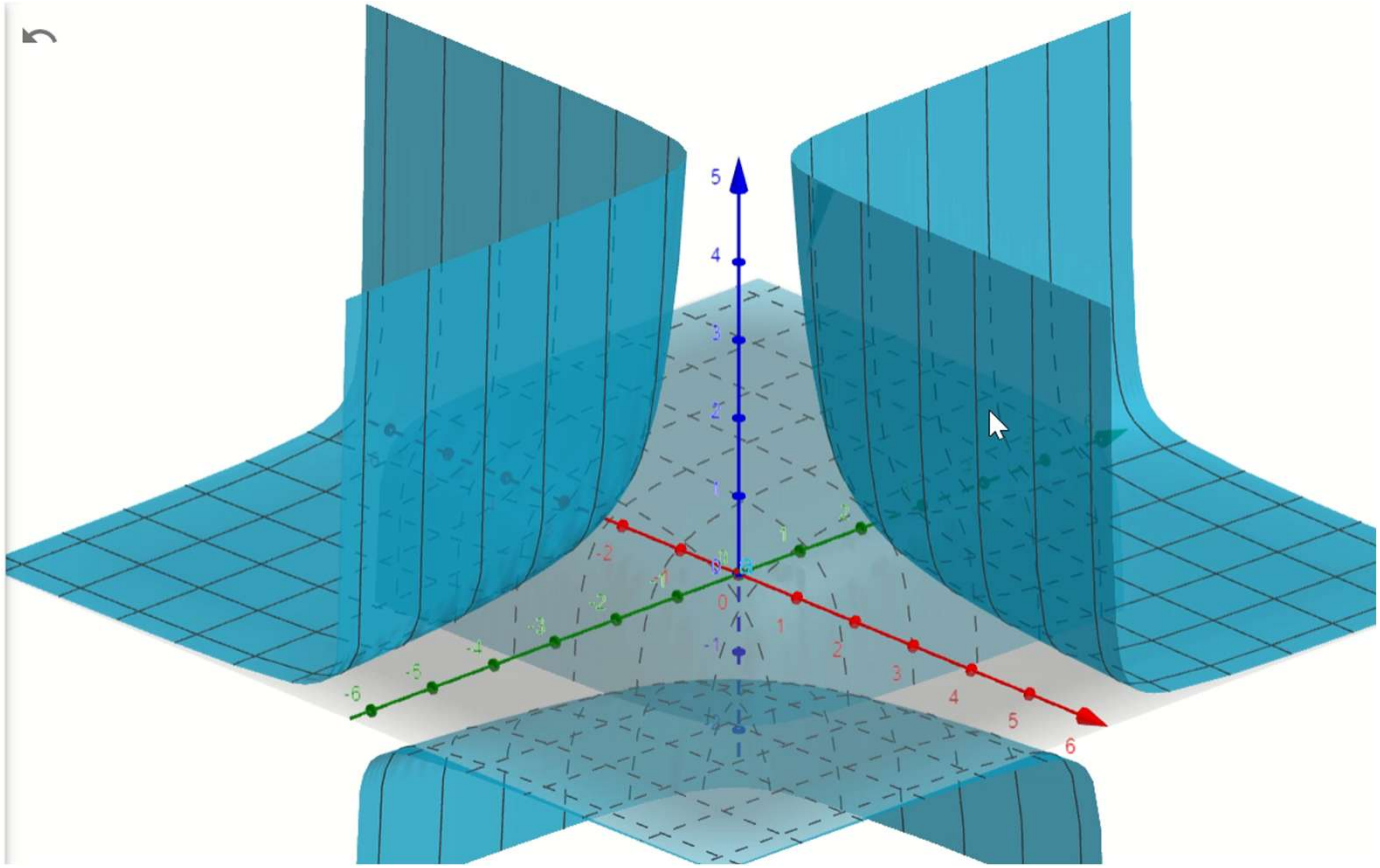
Solution. Define $D_R = \{(x,y) \in \mathbb{R}^2 : R \leq x, y \leq 0.5, 0 < R < 0.5\} \subseteq D$. Then we see that $\iint_D (xy)^{-1}\, dA$ diverges because ($D' = \{(x,y) : 0 < x, y \leq 0.5\}$)

$$\iint_D (xy)^{-1}\, dA \geq \iint_{D'} (xy)^{-1}\, dA = \lim_{R \to 0} \int_R^{0.5} \int_R^{0.5} \tfrac{1}{xy}\, dx dy = \lim_{R \to 0} (\ln(2R))^2 = \infty .$$

$a : z = \frac{1}{x\,y}$

Input...

GeoGebra 3D Calculator

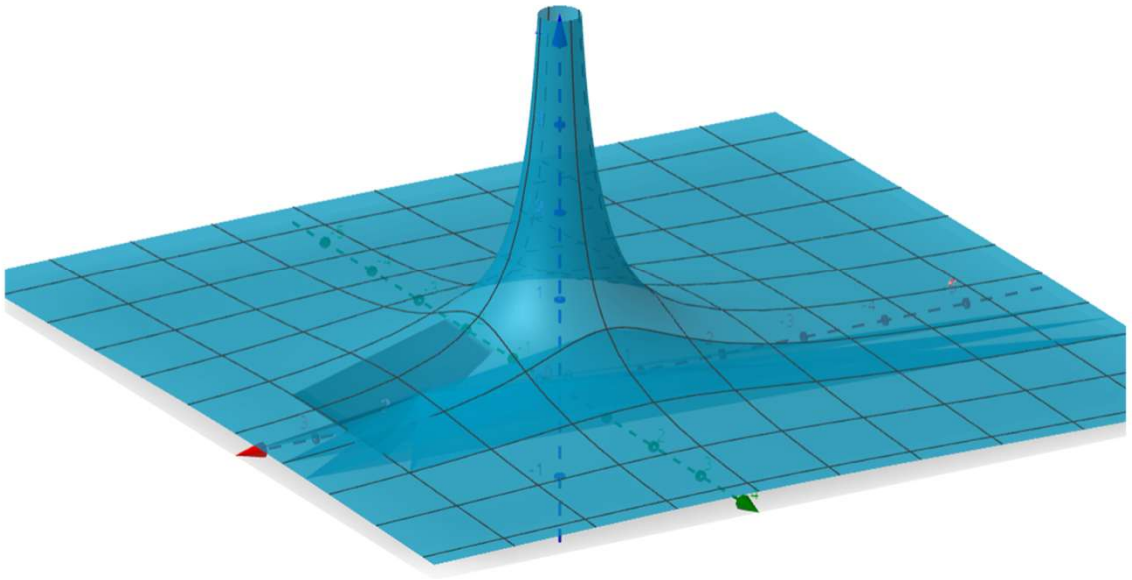


Example 10.11

Find $\iint_D \frac{1}{\sqrt{x^2+y^2}}\,dxdy$, where D is the unit disc centred at $(0, 0)$.

Solution. Denote $D_R = \{(x,y) \in \mathbb{R}^2 : R \le x^2 + y^2 \le 1, 0 < R < 1\} \subset D$ and use the polar coordinates substitution:

$$\iint\limits_D \frac{1}{\sqrt{x^2+y^2}}\,dxdy = \lim_{R\to 0} \iint\limits_{D_R} \frac{1}{\sqrt{x^2+y^2}}\,dxdy = \lim_{R\to 0} \int\limits_0^{2\pi} \int\limits_R^1 \frac{rdrd\theta}{r} = \lim_{R\to 0} 2\pi(1-R) = 2\pi\,.$$

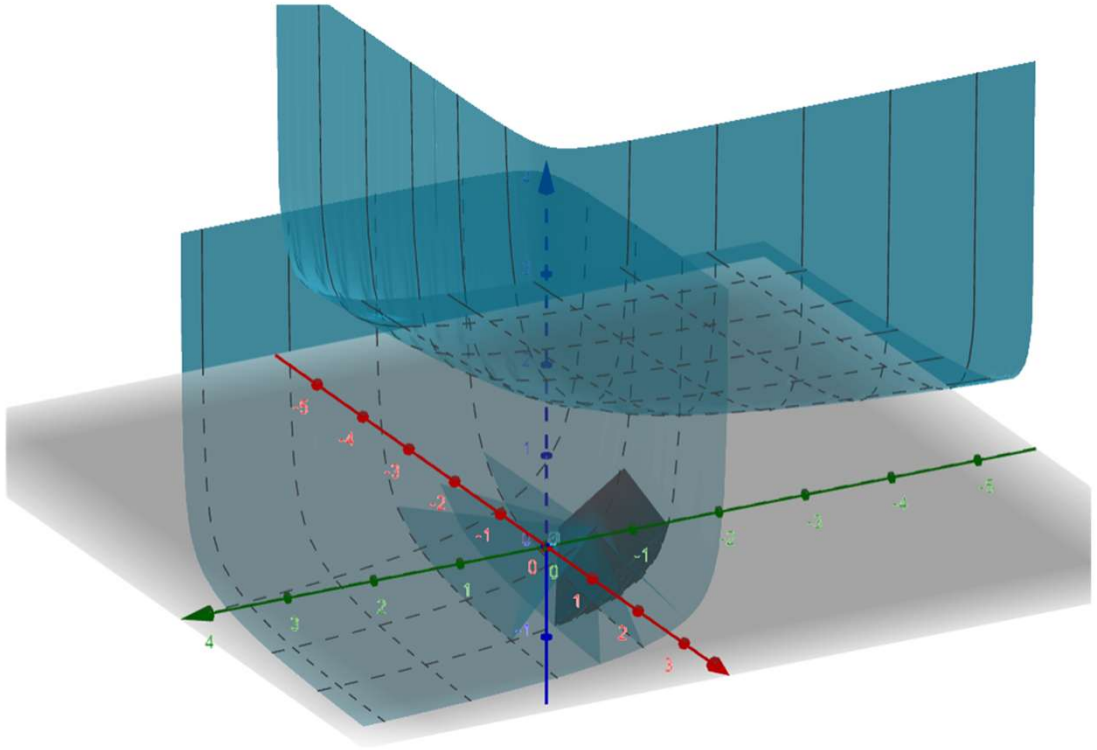


Example 10.12

For $D = \left\{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq 1\right\}$, compute $\iint\limits_D \frac{1}{\sqrt{xy}}\, dxdy$.

Solution. Let $D_{_R} = \{(x, y) : R \leq x, y \leq 1,\ 0 \leq R \leq 1\}$. Then

$$\iint\limits_{D} \frac{1}{\sqrt{xy}}\, dxdy = \lim_{R\to 0} \int\limits_{R}^{1}\int\limits_{R}^{1} \frac{dxdy}{\sqrt{xy}} = \lim_{R\to 0} 2(1-\sqrt{R}) \int\limits_{R}^{1} \frac{1}{\sqrt{y}}\, dy = \lim_{R\to 0} 4(1-\sqrt{R})^2 = 4\,.$$



Example 10.13

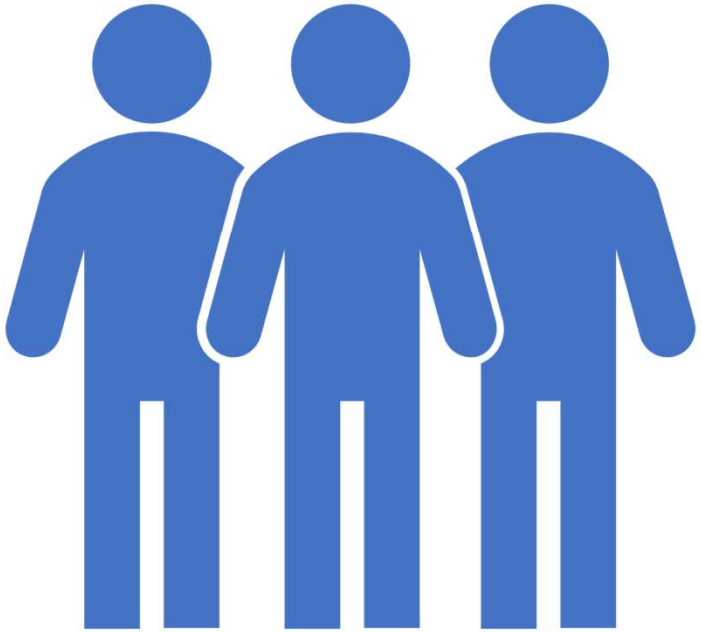
Determine whether the following integral converges $\iint_D \frac{e^{x^2 y}}{xy}\,dxdy$

where $D = \{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq 1\}$.

Solution. Note that $\left|\dfrac{e^{x^2y}}{xy}\right| \geq \dfrac{1}{|x|y}$ and from Example 10.10

$$\iint\limits_{\{(x,y)\in\mathbb{R}^2:0<x,y<1\}} (xy)^{-1}\,dA \text{ is diverging,}$$

so we have $\iint\limits_D \dfrac{e^{x^2y}}{xy}\,dxdy$ also diverging.



That’s all folks!